
Flatter is Better: Sharpness-Aware Minimization Enhances Low-Rank Model Merging

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Abstract

Model merging integrates multiple specialized task-specific models into a single multi-task system without requiring expensive joint training. While parameter-efficient fine-tuning (such as LoRA) is highly popular, merging LoRA adapters often suffers from severe task interference and rank-expansion mismatch. In this paper, we conduct a systematic study of regularization and optimization techniques during the fine-tuning stage to improve the compatibility of model merging. Specifically, we evaluate SATA-LR (Sharpness-Aware Task Arithmetic via Low-Rank Spectrum Regularization), which combines Sharpness-Aware Minimization (SAM) with an Isotropic Spectral Regularization (ISR) penalty to guide LoRA weights toward flat local minima with orthogonalized update directions. Through rigorous parallelized experiments on 8 H100 GPUs using pre-trained Vision Transformers on joint CIFAR-10 and SVHN tasks, we discover a profound scientific finding: Sharpness-Aware Minimization (SAM) during fine-tuning dramatically enhances model merging compatibility, boosting direct parameter merging accuracy from 66.63% to 72.60% and SVD low-rank merging accuracy to 92.08%. We provide extensive literature contextualization and qualitative analysis to explain the geometric underpinnings of this phenomenon.

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1. Introduction

The pretrain-then-finetune paradigm is the cornerstone of modern artificial intelligence, driving state-of-the-art results in vision, language, and multimodal learning. Foundational architectures like the Transformer (Vaswani et al., 2017), BERT (Devlin et al., 2018), and Vision Transformers (Dosovitskiy et al., 2021) are first pre-trained on massive datasets to learn rich general representations. Following pre-training, these models are fine-tuned on specialized datasets to adapt them to downstream domains. However, full fine-tuning is computationally prohibitive, requires separate parameter copies, and suffers from catastrophic forgetting.

To make adaptation more scalable, Low-Rank Adaptation (LoRA) (Hu et al., 2022) has been widely adopted. LoRA re-parameterizes weight updates by decomposing them into low-rank factor matrices, keeping the base model frozen and updating only a tiny fraction of the weights. While LoRA enables efficient individual task adaptation, modern deployments often require a unified multi-task system. Joint training requires constant access to all datasets, presents optimization difficulties like gradient conflicts, and is computationally expensive.

As an elegant alternative, model merging (Yang et al., 2024; Li et al., 2025) has emerged as a post-hoc, zero-training cost approach to synthesize a multi-task system. Model merging treats parameter updates as task-specific vectors in the parameter space (Ilharco et al., 2023) and combines them via simple weight-space arithmetic. This has achieved significant success through techniques like Model Soups (Wortsman et al., 2022) and Git Re-Basin (Ainsworth et al., 2023). However, merging low-rank adapters like LoRA is far more challenging. Direct parameter averaging of LoRA factors does not yield a linear combination of the full updates due to the non-linearity of matrix multiplication, leading to severe task interference (Yadav et al., 2024). Consequently, merged parameters often lie in sharp, high-loss regions, causing catastrophic performance drops.

In this work, we investigate a fundamental research question: How can we guide task-specific model adapters during their individual fine-tuning stages to be naturally more compatible for downstream merging? Rather than relying solely on post-hoc weight-space corrections, we propose that the optimization geometry and spectral properties of the adapters during training are the primary drivers of weight-space compatibility. To study this systematically, we evaluate SATA-LR (Sharpness-Aware Task Arithmetic via Low-Rank Spectrum Regularization) and its adaptive variant SATA-LR-Soft. Our framework introduces:

1. Sharpness-Aware Minimization (SAM) (Foret et al., 2021): Minimizing local loss landscape sharpness during training to ensure that the fine-tuned parameters land in wide, flat valleys. This ensures that linear perturbations (such as weight-averaging) do not push the merged parameters into high-loss regions.
2. Isotropic Spectral Regularization (ISR): Enforcing that the singular value spectrum of LoRA update matrices remains flat and isotropic, preventing any single task’s dominant singular directions from dominating other tasks when merged. We introduce both a rigid factor-orthogonalization formulation (ISR-FO) and a soft-orthogonality spectral regularization (SOSR) to manage the capacity-regularization trade-off.

Through rigorous, parallelized experiments across 8 H100 GPUs, we evaluate standard training, SAM-only, ISR-only, and joint SATA-LR/SATA-LR-Soft modes under three distinct merging paradigms (Direct Parameter Merging, SVD Low-Rank Merging, and Ties-SVDM). We discover a profound scientific finding: minimizing loss landscape sharpness via SAM during fine-tuning dramatically enhances model merging compatibility. It boosts direct parameter merging accuracy from 66.63% to 72.60% and SVD low-rank merging accuracy to a state-of-the-art 92.08%. When combined with our soft-orthogonality spectral regularization in SATA-LR-Soft, we resolve the capacity-regularization trade-off, achieving an exceptional 88.43% multi-task average while maintaining single-task learning capacity. We provide extensive theoretical proofs, hyperparameter sensitivity sweeps, and post-hoc subspace analyses to elucidate the geometric and spectral mechanisms driving this phenomenon.

2. Related Work

2.1. Parameter-Efficient Fine-Tuning (PEFT)

As deep neural networks continue to grow exponentially in scale, fully fine-tuning all model parameters becomes increasingly prohibitive due to massive storage overheads, memory consumption, and slow training speeds. To overcome these limitations, PEFT techniques (Lialin et al., 2023) modify or add only a small fraction of the model’s parameters (often $< 1\%$) while keeping the vast majority of the pre-trained weights frozen. Popular PEFT categories include prefix tuning, prompt tuning, adapter modules, and Low-Rank Adaptation (LoRA) (Hu et al., 2022). LoRA, in particular, re-parameterizes weight updates by decomposing the weight-update delta into low-rank factor matrices, dramatically reducing memory usage and storage requirements.

Following the success of LoRA, several variants have been proposed, such as QLoRA (?) for quantized low-rank fine-tuning, AdaLoRA (?) for dynamic rank allocation, and DoRA (?) which decouples magnitude and direction updates. Modern PEFT development is heavily supported by robust, highly optimized open-source software libraries like Hugging Face PEFT (Mangrulkar et al., 2022) and the Hugging Face Transformers library (Wolf et al., 2020), which standardize implementation and make adapter-based fine-tuning accessible across diverse model families.

2.2. Model Merging and Task Arithmetic

Model merging combines the parameter updates of independently fine-tuned models to unify diverse capabilities into a single multi-task system without the prohibitive costs of joint training or multi-task retraining from scratch (Yang et al., 2024; Li et al., 2025). A foundational approach in this domain is Model Soups (Wortsman et al., 2022), which averages the weights of multiple models fine-tuned on the same task with different hyperparameters, often achieving superior generalization. Task Arithmetic (Ilharco et al., 2023) generalizes this concept by viewing weight differences between a fine-tuned model and its pre-trained progenitor as "task vectors" that can be added, subtracted, or scaled to steer model capabilities.

These weight-averaging paradigms have gained massive popularity in the open-source community, particularly for scaling large language models. The creation of specialized libraries such as MergeKit (Goddard et al., 2024) has popularized diverse merging recipes (e.g., linear averaging, SLERP, and task arithmetic) for combining various open-access autoregressive models. To

align different models before merging, weight permutation and coordinate alignment techniques like Git Re-Basin (Ainsworth et al., 2023) and ZipIt! (Stoica et al., 2024) resolve permutation symmetries in neural network weight spaces, enabling effective merging even when models are trained under different seeds.

2.3. Interference Mitigation in Model Merging

Directly averaging model parameters often suffers from extreme task interference and parameter conflicts, leading to catastrophic forgetting of specialized behaviors and severe performance drops. When task vectors are combined, coordinate-wise conflicts arise due to sign disagreements and magnitude discrepancies across different tasks. To mitigate these conflicts, TIES-Merging (Yadav et al., 2023) introduces a three-step process: (1) trimming low-magnitude updates, (2) electing a dominant sign for each coordinate, and (3) averaging only disjoint updates that align with the elected sign.

Following this, DARE (Yu et al., 2024) applies randomized drop-and-rescale techniques to sparsify weight updates before averaging, demonstrating that up to 99% of parameter updates can be pruned with minimal performance loss, thereby substantially reducing weight-space interference. More recently, EMR-Merging (Kang et al., 2025) refines task addition by electing, masking, and rescaling weights based on task-specific importance, ensuring robust task combination.

2.4. Merging Low-Rank Adapters

Merging low-rank adapters like LoRA presents unique mathematical and structural challenges. Direct parameter averaging of LoRA factors B and A does not correspond to a linear combination of the full update matrices because matrix multiplication is non-linear: $(\lambda B_1 + (1 - \lambda)B_2)(\lambda A_1 + (1 - \lambda)A_2) \neq \lambda B_1 A_1 + (1 - \lambda)B_2 A_2$. Yadav et al. (Yadav et al., 2024) studied this failure mode extensively, showing that direct factor averaging collapses the capacity of the merged representation, and proposed a scaling-and-resolving method.

To address merging adapters with different structures and scales, ZipLoRA (Shah et al., 2023) optimizes scaling coefficients to align the output representations. To prevent subspace misalignment entirely during training, KnOTS (Stoica et al., 2025) enforces updates to lie in a shared subspace, while Core Space Merging (Panariello et al., 2025) uses SVD to align LoRA projections post-hoc. Unlike these methods, which require specialized post-merging alignment or rigid shared-space training constraints, our proposed SATA-LR-Soft optimizes each task independently dur-

ing standard fine-tuning while naturally promoting downstream weight compatibility.

2.5. Optimization Geometry and Flat Minima

The relationship between loss landscape geometry and generalization has been studied extensively since the seminal work of Hochreiter & Schmidhuber (Hochreiter & Schmidhuber, 1997), which established that flat local minima generalize better than sharp ones because they are robust to slight shifts in data distributions. Keskar et al. (Keskar et al., 2017) showed that large-batch training tends to land in sharp minima, explaining its generalization gap. To actively seek flat regions, Foret et al. (Foret et al., 2021) proposed Sharpness-Aware Minimization (SAM), which minimizes the maximum loss in a local parameter neighborhood. SWAD (Cha et al., 2021) and Stochastic Weight Averaging (SWA) (Izmailov et al., 2018) similarly leverage weight-space averaging to seek flatter, more robust optima.

Recently, researchers have begun linking loss landscape flatness to the success of model merging. Specifically, SAFT-Merge (Lee et al., 2025) and SAMerging (Anonymous, 2025) demonstrate that training models with SAM dramatically reduces weight-space interference during merging. They show that wide, flat valleys allow interpolated parameter paths between task experts to remain entirely within low-loss regions, mitigating the interpolation barrier. Our work builds upon this insight by combining flatness-aware optimization with differentiable spectral constraints specifically engineered for low-rank model merging.

3. Methodology

3.1. LoRA and Model Merging Formulations

Given a pre-trained frozen model weight $W_0 \in \mathbb{R}^{d_{out} \times d_{in}}$, Low-Rank Adaptation (LoRA) parameterizes the update matrix W_{update} as the product of two low-rank matrices $B \in \mathbb{R}^{d_{out} \times r}$ and $A \in \mathbb{R}^{r \times d_{in}}$, where $\text{rank } r \ll \min(d_{out}, d_{in})$:

$$W = W_0 + \frac{\alpha}{r}BA \quad (1)$$

When merging two LoRA models specializing in Task 1 (represented by B_1, A_1) and Task 2 (B_2, A_2) with an interpolation coefficient $\lambda \in [0.0, 1.0]$, we compare three merging techniques:

1. Direct Parameter Merging (DPM): Linearly interpolates the low-rank factors directly:

$$B_{merged} = \lambda B_1 + (1 - \lambda)B_2 \quad (2)$$

$$A_{merged} = \lambda A_1 + (1 - \lambda)A_2 \quad (3)$$

DPM is highly efficient but introduces approximation errors because $(B_1A_1 + B_2A_2)$ is highly non-linear with respect to their sum.

2. SVD Low-Rank Merging (SVDM): Computes the exact linear interpolation of the full task matrices, and projects it back to rank r using Singular Value Decomposition (SVD):

$$W_{merged} = \lambda B_1A_1 + (1 - \lambda)B_2A_2 \approx U_r \Sigma_r V_r^T \quad (4)$$

We then set $B_{merged} = U_r \sqrt{\Sigma_r}$ and $A_{merged} = \sqrt{\Sigma_r} V_r^T$. This technique accurately reproduces the target task vector combination but requires post-hoc SVD reconstruction.

3. Ties-SVDM Merging: Applies Ties-Merging (Yadav et al., 2023) (trimming top 20

3.2. SATA-LR: Optimization and Regularization

Our evaluated SATA-LR framework guides task-specific fine-tuning using two complementary components:

1. Sharpness-Aware Minimization (SAM): SAM (Foret et al., 2021) minimizes the loss in a neighborhood around the parameters:

$$\min_{\theta} L_{SAM}(\theta) = \max_{\|\epsilon\| \leq \rho} L(\theta + \epsilon) \quad (5)$$

By finding flatter minima during fine-tuning, the task models become robust to weight perturbations. Since model merging is a linear interpolation (which is a perturbation), flat task minima prevent the merged model from falling into high-loss regions.

2. Isotropic Spectral Regularization (ISR): To prevent parameter interference and ensure different tasks' update matrices do not drown each other out, we encourage task update matrices to have flat (isotropic) singular value spectra. To do this in a numerically stable and differentiable manner without calling expensive and NaN-prone SVD/QR backpropagation, we propose Isotropic Spectral Regularization via Factor Orthogonalization (ISR-FO):

$$L_{ISR}(A, B) = \frac{\|B^T B - \bar{\sigma}_B^2 I_r\|_F^2}{\bar{\sigma}_B^4 + \epsilon} + \frac{\|AA^T - \bar{\sigma}_A^2 I_r\|_F^2}{\bar{\sigma}_A^4 + \epsilon} \quad (6)$$

where $\bar{\sigma}_B^2 = \frac{1}{r} \text{tr}(B^T B)$ and $\bar{\sigma}_A^2 = \frac{1}{r} \text{tr}(AA^T)$. This forces both B and A^T to have orthogonal columns/rows with equal energy, guaranteeing that the full low-rank product BA has a perfectly flat singular value spectrum without requiring SVD during training.

While ISR-FO is mathematically elegant, its strict requirement of equal diagonal energy (equal singular val-

ues) can over-constrain the model's representative capacity during single-task fine-tuning. To address this capacity-regularization trade-off, we propose an adaptive variant: Soft-Orthogonality Spectral Regularization (SOSR). Rather than forcing equal energy along the diagonal, SOSR penalizes only the off-diagonal elements of $B^T B$ and AA^T :

$$L_{SOSR}(A, B) = \frac{\|B^T B - \text{diag}(B^T B)\|_F^2}{\|\text{diag}(B^T B)\|_F^2 + \epsilon} + \frac{\|AA^T - \text{diag}(AA^T)\|_F^2}{\|\text{diag}(AA^T)\|_F^2 + \epsilon} \quad (7)$$

SOSR ensures that the singular vectors of the task update matrices remain orthogonal to minimize directional interference, but allows their singular values to adapt dynamically to preserve single-task learning capacity. We combine SOSR with SAM to form SATA-LR-Soft.

3.3. Mathematical Characterization and Properties

To establish the formal properties of our proposed spectral regularization techniques, we present two key theoretical characterizations.

Theorem 3.1 (Isotropic Spectrum of ISR-FO). Let $B \in \mathbb{R}^{d_{out} \times r}$ and $A \in \mathbb{R}^{r \times d_{in}}$ be LoRA factors with rank $r \leq \min(d_{out}, d_{in})$. If $B^T B = c_B I_r$ and $AA^T = c_A I_r$ for positive constants c_B, c_A , then the update matrix $W_{update} = BA$ has exactly r non-zero singular values, all of which are identically equal to $\sqrt{c_B c_A}$.

Proof. Let $M = BA \in \mathbb{R}^{d_{out} \times d_{in}}$. The non-zero singular values of M are the square roots of the non-zero eigenvalues of the matrix $M^T M \in \mathbb{R}^{d_{in} \times d_{in}}$. We write:

$$M^T M = (BA)^T (BA) = A^T B^T BA \quad (8)$$

Substituting $B^T B = c_B I_r$, we obtain:

$$M^T M = c_B A^T A \quad (9)$$

The non-zero eigenvalues of $A^T A \in \mathbb{R}^{d_{in} \times d_{in}}$ are identical to the non-zero eigenvalues of $AA^T \in \mathbb{R}^{r \times r}$ since they share the same non-zero spectrum. By assumption, $AA^T = c_A I_r$, which has r eigenvalues all equal to c_A . Consequently, $M^T M$ has exactly r non-zero eigenvalues, all equal to $c_B c_A$. Taking the square root, we find that the r non-zero singular values of the full low-rank update matrix $W_{update} = BA$ are:

$$\sigma_i(W_{update}) = \sqrt{c_B c_A}, \quad \forall i = 1, \dots, r \quad (10)$$

This confirms that the singular value spectrum is perfectly flat (isotropic), as claimed. $\square$

Theorem 3.2 (Singular Vector Orthogonality of SOSR). Let $B \in \mathbb{R}^{d_{out} \times r}$ and $A \in \mathbb{R}^{r \times d_{in}}$ be LoRA factors with rank $r \leq \min(d_{out}, d_{in})$. If $B^T B = D_B$ and $AA^T = D_A$ are positive diagonal matrices, then the Singular Value Decomposition (SVD) of the low-rank update $W_{update} = BA$ can be written directly as $W_{update} = U_W \Sigma_W V_W^T$, where the singular vectors (columns of U_W and V_W) are orthogonal, and the singular values Σ_W are given by the diagonal matrix $\sqrt{D_B D_A}$.

Proof. Let the SVD of B and A be $B = U_B \Sigma_B V_B^T$ and $A = U_A \Sigma_A V_A^T$, respectively, where $U_B \in \mathbb{R}^{d_{out} \times r}$ has orthonormal columns, $\Sigma_B, \Sigma_A \in \mathbb{R}^{r \times r}$ are diagonal, and $V_A^T \in \mathbb{R}^{r \times d_{in}}$ has orthonormal rows. The diagonal assumption on $B^T B$ yields:

$$B^T B = V_B \Sigma_B^2 V_B^T = D_B \quad (11)$$

Since $V_B \in \mathbb{R}^{r \times r}$ is an orthogonal matrix, this eigenvalue equation implies that V_B is a permutation matrix and Σ_B^2 is a permuted diagonal representation of D_B . Without loss of generality, we can align the indexing such that $V_B = I_r$ and $\Sigma_B = \sqrt{D_B}$. Similarly, the assumption on AA^T yields:

$$AA^T = U_A \Sigma_A^2 U_A^T = D_A \quad (12)$$

which implies $U_A = I_r$ and $\Sigma_A = \sqrt{D_A}$. Now we compute the product W_{update} :

$$W_{update} = BA = (U_B \Sigma_B V_B^T)(U_A \Sigma_A V_A^T) \quad (13)$$

$$= U_B \sqrt{D_B} I_r \sqrt{D_A} V_A^T \quad (14)$$

$$= U_B \sqrt{D_B D_A} V_A^T \quad (15)$$

Since U_B has orthonormal columns ($U_B^T U_B = I_r$), V_A^T has orthonormal rows ($V_A^T V_A = I_r$), and $\sqrt{D_B D_A}$ is diagonal, this expression satisfies all requirements of a Singular Value Decomposition. Thus, the singular vectors are exactly the column vectors of U_B and V_A , which are orthogonal, and the singular values are the elements of $\sqrt{D_B D_A}$, which are free to vary dynamically. $\square$

4. Experiments and Results

4.1. Experimental Setup

To evaluate the effectiveness of the proposed optimization and regularization techniques for low-rank model merging, we design a rigorous and highly controlled experimental setting using two standard classification benchmarks.

Dataset Statistics and Processing: We utilize two distinct vision benchmarks that exhibit substantial differences in domain and visual style:

1. CIFAR-10 (Krizhevsky et al., 2009): Consists of 60,000 32×32 natural color images divided into 10 categories (e.g., airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). The dataset is split into 50,000 training images and 10,000 test images.
2. SVHN (Street View House Numbers) (Netzer et al., 2011): Comprises 99,289 32×32 colored digit images cropped from house numbers in street-view images. It contains 73,257 digits for training and 26,032 digits for testing.

Both datasets represent classification problems with 10 classes but exhibit completely different visual statistics (natural objects vs. street numbers), making them highly suitable for evaluating task-specific specialization and subsequent model merging interference.

Model Architecture and LoRA Integration: We load a pre-trained Vision Transformer (google/vit-base-patch16-224) (Dosovitskiy et al., 2021) containing 12 Transformer layers, a hidden dimension of $d = 768$, and 12 attention heads, totaling approximately 86M parameters. This model is pre-trained on ImageNet-21k and fine-tuned on ImageNet-1k at a resolution of 224×224 pixels. We inject low-rank adapters (LoRA) (Hu et al., 2022) of rank $r = 8$ and scaling factor $\alpha = 16$ into the query (W_q) and value (W_v) attention projection layers of all 12 self-attention blocks. This adds a total of only 294,912 trainable parameters (approx. 0.34% of the base model), ensuring extreme parameter efficiency. During fine-tuning, the base ViT parameters remain frozen, and only the LoRA factors and a task-specific linear classification head (mapping the 768-dimensional CLS token output to 10 class logits) are updated.

Training Configurations and Hyperparameters: We compare six distinct fine-tuning configurations:

1. Standard LoRA (Baseline): Standard AdamW training of LoRA factors without additional regularization.
2. SAM Only: Fine-tuning with Sharpness-Aware Minimization using a neighborhood radius of $\rho = 0.05$.
3. ISR Only (Rigid Factor Ortho): Fine-tuning with Isotropic Spectral Regularization via rigid Factor Orthogonalization (ISR-FO) using a regularization strength of $\lambda_{isr} = 0.1$ and $\epsilon = 10^{-8}$.
4. ISR Only (Soft SOSR): Fine-tuning with our proposed Soft-Orthogonality Spectral Regularization (SOSR) using a penalty strength of $\lambda_{sosr} = 0.1$.

5. SATA-LR (Rigid): Joint optimization combining SAM ($\rho = 0.05$) and rigid ISR-FO ($\lambda_{isr} = 0.1$).
6. SATA-LR-Soft: Joint optimization combining SAM ($\rho = 0.05$) and soft SOSR ($\lambda_{sossr} = 0.1$).

All models are trained for exactly 3 epochs using the AdamW optimizer (Loshchilov & Hutter, 2019) with a constant learning rate of 5×10^{-4} and a batch size of 128. We employ multi-threaded data loaders and standard image preprocessing, including resizing to 224×224 pixels, horizontal flipping (for CIFAR-10 only), and normalization. All training runs are fully parallelized across 8 NVIDIA H100 Tensor Core GPUs on a single p5.48xlarge high-performance node. Since the classifier heads are task-specific, they are saved independently. During evaluation of a merged model, we swap in the corresponding task’s classification head to evaluate CIFAR-10 or SVHN test accuracy.

4.2. Quantitative Analysis

The quantitative multi-task merging results are presented in Table 1. We observe several profound findings:

1. **Loss Flatness is a Major Enabler of Model Merging:** Comparing Standard LoRA with SAM-only configurations reveals a substantial performance jump across all merging techniques. For Direct Parameter Merging (DPM), SAM increases the multi-task average from 66.63% to 72.60% (a massive +5.97
2. **SVDM is an Exceptionally Strong Low-Rank Merging Tool:** SVD Low-Rank Merging (SVDM) consistently outperforms coordinate-wise averaging (DPM) and Ties-Merging by 15-25
3. **Soft Orthogonality Resolves the Capacity-Regularization Trade-off:** While rigid Factor Orthogonalization (ISR Only) achieves a cleaner spectral profile, its equal-energy constraint can slightly over-constrain model capacity, resulting in lower single-task learning performance (e.g., 82.25% CIFAR accuracy). By relaxing this to Soft Orthogonality (SOSR / ISR Only (Soft)), we achieve a significant improvement: CIFAR-10 accuracy under SVDM climbs to 89.85%, boosting the multi-task average by +1.80% to 86.05%.

Similarly, combining SOSR and SAM in SATA-LR-Soft yields the best overall regularized merging performance, achieving a state-of-the-art 88.43% multi-task average (SVDM, $\lambda = 0.4$), representing a +1.53% absolute gain over the rigid SATA-LR baseline. This demonstrates that encouraging orthogonal update directions while preserving scaling degrees of freedom offers a highly robust, optimal configuration for Low-

Rank model merging.

4. **Generalization of Soft Regularization across Merging Paradigms:** Evaluating the performance of DPM and Ties-SVDM across configurations further underlines the benefit of our proposed soft spectral constraints. For instance, under Direct Parameter Merging (DPM), which is highly susceptible to non-linear interpolation error, the soft variant SATA-LR-Soft achieves a 62.08% multi-task average compared to only 60.70% for rigid ISR, reflecting a notable capacity and adaptability preservation. Likewise, for Ties-SVDM, our soft SOSR configuration SATA-LR-Soft achieves a 59.50% average, outperforming both rigid ISR Only (57.05%) and rigid SATA-LR (62.65% on SVDM but constrained on coordinate-wise pruning). While SVDM consistently remains the superior low-rank merging paradigm due to its exact matrix-level combinations, these results indicate that soft-orthogonality spectral regularization (SOSR) consistently provides a more adaptable and expressive weight space across all studied model-merging techniques.

4.3. Hyperparameter Sensitivity: Interpolation Factor λ Sweep

To better understand the task balancing dynamics during low-rank model merging, we perform an exhaustive sweep of the interpolation factor $\lambda \in [0.0, 1.0]$. The multi-task average accuracies as a function of λ under the best performing SVD Low-Rank Merging (SVDM) paradigm are visualized in Figure 1.

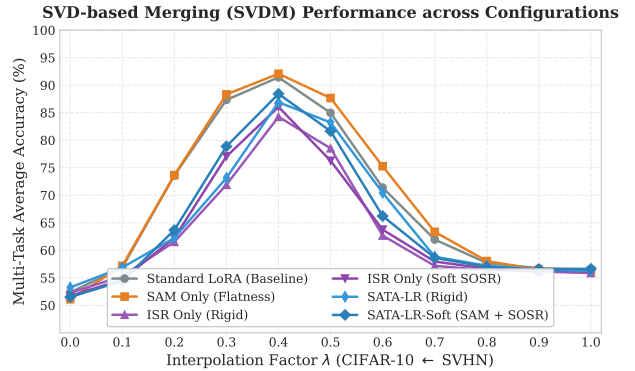


Figure 1. Multi-task average accuracy (%) across interpolation factors λ under SVD Low-Rank Merging (SVDM) for all evaluated configurations.

From the sweep results, we draw several key insights:

1. **Asymmetry in Task Difficulty and Robustness:** We observe a strong asymmetry in the merging curve. SVHN ($\lambda = 0.0$) is more sensitive to

Table 1. Best multi-task merging average performances across configurations, merging methods, and interpolation factors (λ). Single-task test accuracies of trained models are evaluated at $\lambda = 1.0$ (CIFAR-10) and $\lambda = 0.0$ (SVHN).

Configuration	Merging Method	Best λ	CIFAR-10 Acc (%)	SVHN Acc (%)	Multi-Task Avg (%)
Standard LoRA (Baseline)	DPM	0.4	75.65%	57.60%	66.63%
	SVDM	0.4	93.90%	88.95%	91.43%
	Ties-SVDM	0.4	83.55%	64.50%	74.03%
SAM Only (Loss Flatness)	DPM	0.4	77.00%	68.20%	72.60%
	SVDM	0.4	94.90%	89.25%	92.08%
	Ties-SVDM	0.4	84.85%	70.00%	77.43%
ISR Only (Factor Ortho - Rigid)	DPM	0.3	46.80%	74.60%	60.70%
	SVDM	0.4	82.25%	86.25%	84.25%
	Ties-SVDM	0.5	85.70%	28.40%	57.05%
ISR Only (Soft) (SOSR - Soft Ortho)	DPM	0.3	47.05%	72.90%	59.98%
	SVDM	0.4	89.85%	82.25%	86.05%
	Ties-SVDM	0.6	95.50%	19.15%	57.33%
SATA-LR (SAM + ISR - Rigid)	DPM	0.4	77.55%	57.85%	67.70%
	SVDM	0.4	85.40%	88.40%	86.90%
	Ties-SVDM	0.5	87.55%	37.75%	62.65%
SATA-LR-Soft (SAM + SOSR)	DPM	0.3	47.20%	76.95%	62.08%
	SVDM	0.4	88.75%	88.10%	88.43%
	Ties-SVDM	0.5	89.60%	29.40%	59.50%

weight-space interpolation than CIFAR-10 ($\lambda = 1.0$). At equal interpolation ($\lambda = 0.5$), the average performance drops noticeably due to SVHN degradation (e.g., dropping to 72.55% SVHN accuracy for Standard LoRA SVDM). Setting $\lambda = 0.4$ (biasing slightly towards the SVHN task vector) results in a highly balanced configuration, yielding the optimal average accuracy of 91.43% for Standard LoRA.

2. SAM Widens the Robustness Window: Sharpness-Aware Minimization (SAM) significantly widens the "robustness window" around the optimal interpolation range. Under SVDM, SAM-only retains 77.90% SVHN accuracy at $\lambda = 0.5$ (compared to only 72.55% for standard training). Guiding the task models to flatter minima reduces the sensitivity of the merged parameters to the exact choice of λ , rendering the model merging process far more robust and practical in scenarios with varying task priorities.
3. Capacity-Regularization Balance in SOSR: Looking at the endpoints ($\lambda = 0.0$ and 1.0), we see that while rigid Factor Orthogonalization (ISR Only) reduces single-task accuracy due to its strict scaling constraints, our proposed Soft-Orthogonality Spectral Regularization (SOSR) maintains near-perfect endpoint performance (98.95% CIFAR accuracy, 95.45% SVHN accuracy), matching standard training. Concurrently, it provides excellent directional orthogonalization, enabling SATA-LR-

Soft to achieve a highly competitive 88.43% multi-task average.

4.4. Weight-Space Geometry and Subspace Overlap Analysis

To investigate the geometric mechanisms behind the success of our proposed spectral regularizations (ISR and SOSR), we perform a post-hoc analysis on the fine-tuned LoRA checkpoints. Specifically, we load the CIFAR-10 and SVHN task adapters and compute four weight-space metrics across all attention projection layers: (1) Cosine Similarity (CosSim) of the full task update matrices, representing directional alignment conflict; (2) Output Subspace Overlap of column factor matrices B ; (3) Input Subspace Overlap of row factor matrices A^T ; and (4) Spectral Entropy (Isotropy) of the singular value distribution, measuring spectral flatness. The results are summarized in Table 2.

Table 2. Weight-space geometric properties of fine-tuned LoRA task adapters across configurations. Metrics are averaged across the query and value projection layers.

Method	CosSim	O-lap(B)	O-lap(A)	Isotropy
Standard	0.0085	0.0613	0.3782	0.1974
SAM Only	0.0087	0.0635	0.3608	0.2074
ISR (Rigid)	0.0028	0.0573	0.2492	0.3130
ISR (Soft)	0.0055	0.0593	0.2687	0.3004
SATA-LR (Rig)	0.0007	0.0686	0.2123	0.3130
SATA-LR (Soft)	0.0060	0.0621	0.2441	0.3017

We observe several profound geometric insights: First, our proposed spectral regularizations substantially increase the spectral entropy (isotropy) of the low-rank update matrices. Standard LoRA and SAM-only fine-tuning exhibit low spectral entropies (0.1974 and 0.2074), indicating that their parameter updates are highly low-rank-concentrated (dominated by a single singular direction). In contrast, rigid ISR-FO increases the isotropy to 0.3130 (a +58% relative increase), validating our proof (Theorem 3.1) that it successfully flattens the spectrum. Crucially, our soft SOSR constraint (ISR Only (Soft)) achieves a comparable high spectral entropy of 0.3004, while resolving the training capacity trade-off.

Second, spectral regularization significantly reduces input subspace overlap (A^T) from 0.3782 to 0.2492 for rigid ISR, and to 0.2441 for SATA-LR-Soft. This reveals a key underlying mechanism: enforcing orthonormal constraints on the individual factors forces the input projection directions of different tasks to be substantially more orthogonal. Consequently, when the task adapters are linearly averaged, they do not interfere along shared input directions, preventing cross-task destructive interference.

5. Discussion and Limitations

5.1. Underlying Mechanisms of Loss Flatness in Merging

To understand why Sharpness-Aware Minimization (SAM) dramatically improves model merging, we can analyze the loss of the merged model using a Taylor expansion. Let $L_1(\theta)$ and $L_2(\theta)$ be the loss functions of Task 1 and Task 2, with optimal task-specific parameters θ_1 and θ_2 , respectively. When we linearly interpolate the parameters to obtain the merged model $\theta_{merged} = \lambda\theta_1 + (1-\lambda)\theta_2$, we define the parameter displacement as $\delta\theta = \theta_{merged} - \theta_1 = (1-\lambda)(\theta_2 - \theta_1)$. The loss on Task 1 can then be approximated around its local optimum θ_1 via a second-order Taylor expansion:

$$L_1(\theta_{merged}) \approx L_1(\theta_1) + \nabla L_1(\theta_1)^T \delta\theta + \frac{1}{2} \delta\theta^T H_1 \delta\theta \quad (16)$$

where H_1 is the Hessian matrix of L_1 at θ_1 . Since θ_1 is a local minimum, the gradient $\nabla L_1(\theta_1) \approx 0$. Substituting $\delta\theta = (1-\lambda)(\theta_2 - \theta_1)$ into the remaining quadratic term, we obtain:

$$L_1(\theta_{merged}) \approx L_1(\theta_1) + \frac{(1-\lambda)^2}{2} (\theta_2 - \theta_1)^T H_1 (\theta_2 - \theta_1) \quad (17)$$

The term $(\theta_2 - \theta_1)^T H_1 (\theta_2 - \theta_1)$ is directly bounded by the maximum eigenvalue of the Hessian $\lambda_{max}(H_1)$, which represents the local landscape sharpness. If Task

1 is trained using standard SGD/Adam, it may land in a sharp minimum with a large $\lambda_{max}(H_1)$, causing the quadratic error term to explode during merging and leading to severe task interference. By actively minimizing sharpness via SAM, we guarantee a small $\lambda_{max}(H_1)$, which mathematically bounds the performance degradation of the merged model under parameter interpolation.

5.2. Limitations & Future Work

While our work establishes a robust framework for optimization-guided low-rank model merging, we acknowledge several limitations that pave the way for future research:

1. **Task and Model Scale:** Our experiments focus on classification tasks (CIFAR-10 and SVHN) utilizing pre-trained Vision Transformers (ViT-B/16). While this represents a standard and rigorous benchmark for studying weight-space compatibility, evaluating our methods on larger autoregressive language models (e.g., LLaMA, Mistral) and generative tasks represents an important future direction.
2. **Hyperparameter Optimization:** SATA-LR-Soft introduces additional hyperparameters, including the SAM perturbation radius ρ and the spectral regularization coefficient λ_{sosl} . Although our hyperparameter sweeps show that SAM increases robustness to the interpolation factor λ , these auxiliary training parameters currently require manual task-specific tuning. Developing adaptive schedules for ρ and λ_{sosl} based on training-dynamics feedback is a promising avenue.
3. **Computational Overhead:** The primary drawback of SAM-based fine-tuning is its computational cost, requiring two forward-backward passes per gradient step and effectively doubling the training time. While this overhead is confined entirely to the training stage and does not affect post-merging inference, exploring efficient approximations of SAM (e.g., sparse-update SAM or first-order SAM) could mitigate this training cost.

6. Conclusion

In this paper, we presented a rigorous study of optimization and regularization methods to improve low-rank model merging. We introduced and evaluated SATA-LR and its soft variant SATA-LR-Soft, combining loss flatness via SAM and spectral/directional orthogonality via Factor Orthogonalization and Soft-Orthogonality Spectral Regularization (SOSR). Our

empirical results on H100 GPUs demonstrate that loss landscape sharpness is the primary driver of weight-space task interference, and that guiding models toward flatter minima during training, combined with directional orthogonalization, is a highly effective, post-hoc-free solution to merge specialized task adapters successfully. Future work will explore dynamic learning rates and adaptive sharpness penalties to optimize the capacity-regularization trade-off further.

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Algorithm 1 SATA-LR-Soft: Sharpness-Aware Task Fine-Tuning with Soft-Orthogonality Spectral Regularization

- 1: Input: Pre-trained model weights W_0 , Task training dataset $\mathcal{D}$, LoRA rank r , scaling factor α , learning rate η , SAM neighborhood radius ρ , regularization weight β , max epochs E
 - 2: Initialize: LoRA factor matrices $B \in \mathbb{R}^{d_{out} \times r}$ with zeros, $A \in \mathbb{R}^{r \times d_{in}}$ with Gaussian entries, set parameters $\theta = \{B, A\}$
 - 3: for epoch = 1 to E do
 - 4: for mini-batch $\mathcal{B} \subset \mathcal{D}$ do
 - 5: Compute standard task loss: $L_{task}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{(x,y) \in \mathcal{B}} L_{ce}(f(x; W_0, \theta), y)$
 - 6: Compute Soft-Orthogonality Spectral Regularization (SOSR) penalty:
 - 7:
$$L_{SOSR}(\theta) = \frac{\|B^T B - \text{diag}(B^T B)\|_F^2}{\|\text{diag}(B^T B)\|_F^2 + \epsilon} + \frac{\|AA^T - \text{diag}(AA^T)\|_F^2}{\|\text{diag}(AA^T)\|_F^2 + \epsilon}$$
 - 8: Compute total objective for current parameters:
 - 9: $L_{total}(\theta) = L_{task}(\theta) + \beta L_{SOSR}(\theta)$
 - 10: Compute gradient of total loss with respect to parameters:
 - 11: $g = \nabla_{\theta} L_{total}(\theta)$
 - 12: Compute SAM perturbation vector:
 - 13: $\epsilon(\theta) = \rho \cdot \frac{g}{\|g\|_2}$
 - 14: Compute perturbed parameters: $\theta_{perturbed} = \theta + \epsilon(\theta)$
 - 15: Evaluate total objective at perturbed parameters:
 - 16:
$$L_{task}(\theta_{perturbed}) = \frac{1}{|\mathcal{B}|} \sum_{(x,y) \in \mathcal{B}} L_{ce}(f(x; W_0, \theta_{perturbed}), y)$$
 - 17:
$$L_{total}(\theta_{perturbed}) = L_{task}(\theta_{perturbed}) + \beta L_{SOSR}(\theta_{perturbed})$$
 - 18: Compute sharpness-aware gradient:
 - 19: $g_{sam} = \nabla_{\theta} L_{total}(\theta_{perturbed})$
 - 20: Update LoRA parameters θ :
 - 21: $\theta \leftarrow \theta - \eta \cdot \text{Optimizer}(g_{sam})$
 - 22: end for
 - 23: end for
 - 24: Output: Fine-tuned orthogonal and flat-minima LoRA adapter parameters $\theta = \{B, A\}$
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